RISHI THAKUR

DATA SCIENTIST

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PROFESSIONAL SUMMARY

Data Scientist with a quantitative background (M.Tech CSE, GATE CS qualified) and expertise in exploratory data analysis (EDA), statistical inference, machine learning, and predictive modeling. Experienced handling class imbalances via SMOTE + Tomek Links and deploying regression and classification pipelines. Skilled at translating operational bottlenecks into statistical models and executive dashboards.

SKILLS

- **Machine Learning & Statistics:** Machine Learning (Random Forest, XGBoost, KMeans, Regression), Feature Engineering, Resampling (SMOTE + Tomek Links), Hypothesis Testing, Statistical Inference, R^2, RMSE, F1-Score
- **Deep Learning & NLP:** Diffusion Models (DDPM/DDIM), Recurrent Encoders (GRU), Transformers, Multi-Head Attention, BERT Embeddings, PyTorch, TensorFlow, Scikit-learn
- **Data Systems & Visualization:** Python (Pandas, NumPy), Advanced SQL (Joins, CTEs, Window Functions), MySQL, MariaDB, OracleDB, Microsoft Power BI (DAX), Linux/Bash, Git
- **Communication & Spoken Skills:** Executive Data Storytelling, Statistical Interpretation for Non-Technical Stakeholders, Research Presentations, Cross-Functional Collaboration, Technical Writing

WORK EXPERIENCE

Chhattisgarh Infotech Promotion Society (CHiPS) | Raipur, India | 06/2025 – 03/2026
Data Science & Analytics Intern • Dept. of Health & Family Welfare, Govt. Of Chhattisgarh

- Conducted exploratory data analysis (EDA) across **33 districts** on healthcare attendance records, addressing schema anomalies and duplicate designations.
- Engineered an automated **SQL/ETL feature pipeline** mapping AEBAS biometric IDs to State Treasury payroll codes, establishing cross-departmental ground-truth records.
- Formulated a **3-tier risk classification model** tracking bonded medical personnel across facilities, pinpointing critical compliance gaps.
- Reconciled disparate database schemas across MariaDB, State Treasury, and e-Hospital systems with **100% data fidelity**.
- Architected an executive 6-page **Power BI dashboard** featuring **4-tier drill-down hierarchies** (District → Facility → Designation → Individual) to present core KPIs.

KEY PROJECTS

Proactive 5G Handover Prediction (Diffusion Models) • *M.Tech Thesis — Predictive Modeling & Telemetry*
Technologies: Python, PyTorch, GRU, DDPM/DDIM, Scikit-learn, Random Forest, SMOTE, Tomek Links

- Architected a predictive engine pairing a **GRU encoder** with a **Diffusion Model (DDPM/DDIM)** forecasting signal degradation from multivariate **spatio-temporal** telemetry.
- Resolved **98%/2% class imbalance** using **SMOTE + Tomek Links**, training a Random Forest classifier to attain a **97.64% F1-score**.

AI vs. Human Text Classification Pipeline • *Applied NLP & Deep Learning*
Technologies: Python, BERT, XGBoost, Bi-LSTM, Transformers, 1D-CNN, TensorFlow, Scikit-learn

- Extracted 768-dim dense **BERT embeddings** paired with **XGBoost**, and constructed a deep hybrid Bi-LSTM/Transformer network.
- Applied stratified sampling (80/20), text normalization (NFKD), dropout (0.5), and dynamic learning rate decay, achieving **99% Precision** and **98% Recall** in synthetic LLM classification.

EDUCATION

M.Tech in Computer Science & Engineering | IIIT Naya Raipur (CGPA: 7.90) | 08/2024 – 07/2026
B.Tech in Computer Science | Amity University Chhattisgarh (CGPA: 7.71) | 08/2019 – 06/2023

CERTIFICATIONS & KEY QUALIFICATIONS

- **Qualified GATE Computer Science (CS)**, 2024
- **Python for Data Science & AI** (Coursera)
- **Celonis Process Mining Foundations** (Credly)
- **Microsoft Power BI** (Udemy)
- **SQL & Database Design** (Udemy)
- **IELTS Academic:** 7.5 Band Score